Evaluation runs: **bcast** (broadcast-consensus-civl)

generated 2026-11-08 from brain-data 38584f536 2026-11-08 20:45:11 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	8	16	conforms	0.0	4	1 h, 250 GB	4	16	brain03, 2026-11-01
3	48	1728	conforms	0.1	2	1 h, 250 GB	8	172	brain03, 2026-11-01
4	384	5308416	conforms	32.3	26	48 h, 250 GB	16	2684	brain05, 2026-11-07
5			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
2	8	16	14	violation	0.0	2	1 h, 250 GB	16	120	brain03, 2026-11-01
3	48	1728	1688	violation	3.8	2	48 h, 250 GB	1693	81153	brain01, 2026-11-07
4	384			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early, 5 bag decide-early.t16.p.r3, 5 bag correct.t16.p.r3, 6 bag decide-early, 6 bag correct.

left out by the table rules (ok): 3 bag decide-early, 4 bag correct.

skipped: a smaller size had reached its limit: 5 decide-early bag, 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2	8	16	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-01
3	48	1728	conforms	0.1	4	1 h, 250 GB	brain02, 2026-11-01
4	384	5308416	conforms	29.1	25	48 h, 250 GB	brain05, 2026-11-07
5			🕓 (48 h)			48 h, 250 GB	brain05, 2026-11-02

stateful: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2	8	16	14	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-01
3	48	1728	1688	violation	0.6	2	48 h, 250 GB	brain01, 2026-11-07
4	384	5308416	5292864	violation	29.6	25	48 h, 250 GB	brain05, 2026-11-03
5				(48 h)			48 h, 250 GB	brain05, 2026-11-03
7				(48 h)			48 h, 250 GB	brain02, 2026-11-05

Not run (stateful). planned but never scheduled (grid points beyond the ladder): 4 bag correct.t16.r31, 4 bag correct.t16.r32, 4 bag correct.t16.r33, 4 bag correct.t16.r34, 4 bag correct.t16.r35, 5 bag decide-early.t16.p.r3, 5 bag correct.t16.p.r3, 7 decide-early bag, 7 correct bag, 8 decide-early bag, 8 correct bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag, 11 correct bag, 11 decide-early bag, 12 decide-early bag, 12 correct bag.

left out by the table rules (ok): 3 bag decide-early, 4 bag correct, 4 bag correct.t16.r41, 4 bag correct.t16.r42, 4 bag correct.t16.r43, 4 bag correct.t16.r44, 4 bag correct.t16.r45.

left out by the table rules (timeout): 5 correct bag, 5 decide-early bag, 6 bag correct, 6 bag decide-early.

skipped: a smaller size had reached its limit: 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

the tool ended with an error (exit-101): 7 bag correct.

left out by the table rules (invalid: foreign load): 8 bag correct, 8 bag decide-early.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	conforms	0.0	8	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	48	1728	conforms	0.1	8	1 h, 250 GB	0.0	0.1	0.0	merge	brain05, 2026-11-03
4	384	5308416	conforms	34.4	25	48 h, 250 GB	0.1	34.4	0.0	merge	brain05, 2026-11-07

offline: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	14	violation	0.0	8	1 h, 250 GB	0.0	0.0	0.0	merge	brain05, 2026-11-03
3	48	1728	211	violation	0.2	8	48 h, 250 GB	0.1	0.2	0.0	merge	brain01, 2026-11-07
4	384	5308416	4084	violation	28.7	29	1 h, 250 GB	0.1	28.5	0.2	merge	brain05, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): 4 bag correct.r31, 4 bag correct.r32, 4 bag correct.r33, 4 bag correct.r34, 4 bag correct.r35, 5 bag decide-early.p.r3, 5 bag correct.p.r3, 6 bag decide-early, 6 bag correct.

left out by the table rules (ok): 3 bag decide-early, 4 bag correct, 4 bag correct.r41, 4 bag correct.r42, 4 bag correct.r43, 4 bag correct.r44, 4 bag correct.r45.

skipped: a smaller size had reached its limit: 5 correct bag, 5 decide-early bag, 5 bag correct, 5 bag decide-early, 6 decide-early bag, 6 correct bag, 7 decide-early bag, 7 correct bag, 8 correct bag, 8 decide-early bag, 9 correct bag, 9 decide-early bag, 10 decide-early bag, 10 correct bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	conforms	4.2	168	48 h, 250 GB	0.6	4.2	0.0	merge	brain01, 2026-11-07
3	48	1728	conforms	8.0	312	48 h, 250 GB	0.1	8.0	0.0	merge	brain01, 2026-11-07

cross offline, Java implementation against the Rust specification: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	14	violation	4.1	158	48 h, 250 GB	0.6	4.1	0.0	merge	brain01, 2026-11-07
3	48	1728	211	violation	9.5	386	48 h, 250 GB	0.1	9.4	0.1	merge	brain01, 2026-11-07
4	384	5308416	4084	violation	11515.1	72399	48 h, 250 GB	0.1	11515.0	0.1	merge	brain02, 2026-11-08

Not run (cross offline, Java implementation against the Rust specification). skipped: a smaller size had reached its limit: 4 bag correct, 6 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early, 6 bag correct.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	conforms	3.9	156	10 min, 250 GB	3.9	0.0	0.0	merge	brain01, 2026-11-03
3	48	1728	conforms	4.1	153	10 min, 250 GB	4.1	0.1	0.0	merge	brain01, 2026-11-03
4	384	5308416	conforms	28.9	185	10 min, 250 GB	4.9	28.7	0.2	merge	brain01, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	8	16	14	violation	3.9	156	10 min, 250 GB	3.9	0.0	0.0	merge	brain01, 2026-11-03
3	48	1728	211	violation	4.3	153	10 min, 250 GB	4.1	0.1	0.2	merge	brain01, 2026-11-03
4	384	5308416	4084	violation	27.7	187	10 min, 250 GB	4.9	27.5	0.2	merge	brain01, 2026-11-03

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early, 6 bag correct, 6 bag decide-early.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	16	conforms	0.8	150	1 h, 250 GB		brain05, 2026-11-03
3	48	1728	conforms	6.0	447	1 h, 250 GB		brain05, 2026-11-03
4	384		(1 h)			1 h, 250 GB		brain05, 2026-11-03

cross online, Java implementation (JMC) against the Rust specification export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	16	14	violation	0.8	150	1 h, 250 GB	48	brain05, 2026-11-03
3	48	1728	1688	violation	5.0	403	1 h, 250 GB	54	brain05, 2026-11-03
4	384			(1 h)			1 h, 250 GB		brain05, 2026-11-03

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 2 bag decide-early, 2 bag correct, 3 bag decide-early, 3 bag correct, 4 bag decide-early, 4 bag correct, 5 bag decide-early, 5 bag correct, 6 bag decide-early, 6 bag correct.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	8	16	conforms	3.4	171	1 h, 250 GB		brain01, 2026-11-02
3	48	1728	conforms	3.5	155	1 h, 250 GB		brain01, 2026-11-02
4	384	5308416	conforms	37.5	171	1 h, 250 GB		brain01, 2026-11-02
6	46080		(48 h)			48 h, 200 GB		volta15, 2026-11-05

cross online, Rust implementation (TraceForge) against the Java specification export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	8	16	14	violation	3.4	171	1 h, 250 GB	1	brain01, 2026-11-02
3	48	1728	1688	violation	3.5	155	1 h, 250 GB	2	brain01, 2026-11-02
4	384	5308416	5292864	violation	35.5	171	1 h, 250 GB	2	brain01, 2026-11-02

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early, 6 bag decide-early.

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.3	170	10 min, 250 GB	4.2	0.0	0.1	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.0	312	10 min, 250 GB	8.0	0.1	0.0	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.1	160	10 min, 250 GB	4.1	0.0	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	9.6	385	10 min, 250 GB	9.4	0.1	0.2	merge	brain01, 2026-11-03

twins offline, Rust implementation against its Java twin: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.2	170	10 min, 250 GB	4.2	0.0	0.0	merge	brain01, 2026-11-03
3	1728	1728	211	violation	8.2	312	10 min, 250 GB	8.0	0.1	0.2	merge	brain01, 2026-11-03

Not run (twins offline, Rust implementation against its Java twin). refused (an input missing at run time): 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.2	170	10 min, 250 GB	0.0	4.2	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.0	312	10 min, 250 GB	0.1	8.0	0.0	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	4.9	160	10 min, 250 GB	0.0	4.1	0.8	merge	brain01, 2026-11-03
3	1728	1728	conforms	9.9	385	10 min, 250 GB	0.1	9.4	0.5	merge	brain01, 2026-11-03

twins offline, Java implementation against its Rust twin: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.2	160	10 min, 250 GB	0.0	4.1	0.0	merge	brain01, 2026-11-03
3	1728	1728	211	violation	9.4	386	10 min, 250 GB	0.1	9.4	0.0	merge	brain01, 2026-11-03

Not run (twins offline, Java implementation against its Rust twin). skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	6.1	169	10 min, 250 GB	6.0	0.0	0.1	merge	brain01, 2026-11-03
3	1728	1728	conforms	11.0	2407	10 min, 250 GB	10.9	0.1	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	3.9	156	10 min, 250 GB	3.7	0.0	0.2	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.2	376	10 min, 250 GB	8.2	0.1	0.1	merge	brain01, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	6.2	169	10 min, 250 GB	6.0	0.0	0.2	merge	brain01, 2026-11-03
3	1728	1728	211	violation	10.9	2407	10 min, 250 GB	10.9	0.1	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Rust implementation against the context-free Java port). refused (an input missing at run time): 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	6.2	169	10 min, 250 GB	0.0	6.0	0.1	merge	brain01, 2026-11-03
3	1728	1728	conforms	11.0	2407	10 min, 250 GB	0.1	10.9	0.1	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	conforms	3.7	156	10 min, 250 GB	0.0	3.7	0.0	merge	brain01, 2026-11-03
3	1728	1728	conforms	8.2	376	10 min, 250 GB	0.1	8.2	0.0	merge	brain01, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2	16	16	14	violation	4.1	156	10 min, 250 GB	0.0	3.7	0.4	merge	brain01, 2026-11-03
3	1728	1728	211	violation	8.3	377	10 min, 250 GB	0.1	8.2	0.1	merge	brain01, 2026-11-03

Not run (context-free twins offline, Java port against the Rust implementation). skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.

planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	6.4	167	1 h, 250 GB		brain01, 2026-11-03
3	1728	1728	conforms	13.0	2266	1 h, 250 GB		brain01, 2026-11-03
4	5308416		🕐 (1 h)			1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	6.1	157	1 h, 250 GB		brain01, 2026-11-03
3	1728	1728	conforms	13.4	2435	1 h, 250 GB		brain01, 2026-11-03
4	5308416		🕐 (1 h)			1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	16	16	14	violation	6.5	154	1 h, 250 GB	174	brain01, 2026-11-03
3	1728	1728	1688	violation	12.7	2332	1 h, 250 GB	137	brain01, 2026-11-03
4	5308416			(1 h)			1 h, 250 GB		brain01, 2026-11-02

Not run (context-free twins online, Java port (JMC) against the Rust implementation's export). planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	3.2	156	1 h, 250 GB		brain01, 2026-11-02
3	1728	1728	conforms	6.9	465	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **decide-early**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2	16	16	conforms	3.2	172	1 h, 250 GB		brain01, 2026-11-02
3	1728	1728	conforms	6.6	437	1 h, 250 GB		brain01, 2026-11-02

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair **decide-early**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	16	16	14	violation	3.2	156	1 h, 250 GB	1	brain01, 2026-11-02
3	1728	1728	1688	violation	6.9	465	1 h, 250 GB	2	brain01, 2026-11-02

Not run (context-free twins online, Rust implementation (TraceForge) against the Java port's export). skipped: a smaller size had reached its limit: 4 bag correct, 4 bag decide-early.
planned but never scheduled (grid points beyond the ladder): 5 bag correct, 5 bag decide-early.